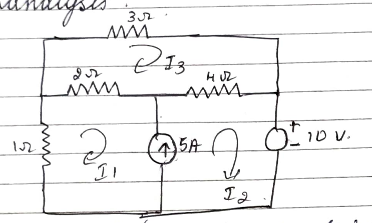
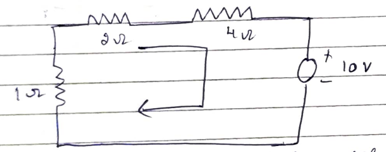
Module (01): - Basic circuit analysis concepts

Unit 1: -

① Determine the loop currents using Mesh analysis.



=> Mesh (01) and (02) forms supermesh



Applying KVL to supermesh

$-I_1 - 2(I_1 - I_3) - 4(I_2 - I_3) = 10$

$\boxed{-3I_1 - 4I_2 + 6I_3 = 10}$ —— ①

constraint Eqn.

$I_2 - I_1 = 5$ —— ②

Apply KVL to loop 3

$-3I_3 - 4(I_3 - I_2) - 2(I_3 - I_1) = 0$

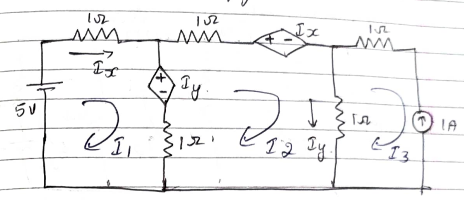
$+2I_1 + 4I_2 - 9I_3 = 0$

$-2I_1 - 4I_2 + 9I_3 = 0$ ← ③

solving ①, ②, ③

$\boxed{I_1 = -5.5A}$ $\boxed{I_2 = -0.55A}$ $\boxed{I_3 = -1.48A}$

2) Find the current in the three meshes for the figure shown.



=> $I_x = I_1$

$I_y = I_2 - I_3$

Apply KVL for mesh 01

$5 - I_1 - I_y - 1(I_1 - I_2) = 0$

$5 - I_1 - I_y - I_1 + I_2 = 0$

$5 - I_1 - (I_2 - I_3) - I_1 + I_2 = 0$

$5 - I_1 - \cancel{I_2} + I_3 - I_1 + \cancel{I_2} = 0$

$-2I_1 + I_3 = -5$ —— ①

Apply KVL to mesh 02

$-I_2 - I_x - 1(I_2 - I_3) - 1(I_2 - I_1) + I_y = 0$

$-\cancel{I_2} - \cancel{(I_1)} - I_2 + \cancel{I_3} - I_2 + \cancel{I_1} + \cancel{I_2} - \cancel{I_3} = 0$

$-2I_2 = 0$

$\boxed{I_2 = 0A}$ —— ②

From mesh 03

$\boxed{I_3 = 1A}$

substitute in ①

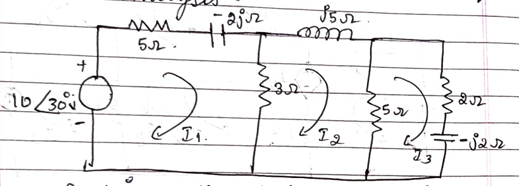
~~$-2I_1 + I_3 = -5$~~ $-2I_1 + I_3 = -5$

~~$-2I_1 + 3 = -5$~~ $-2I_1 + 1 = -5$

$-2I_1 = -4$

$\boxed{I_1 = 2A}$

③ Determine loop currents using mesh analysis.



=) Applying KVL to mesh (01).

$10\angle 30° - 5I_1 - 2jI_1 - 3(I_1 - I_2) = 0$

$10\angle 30 - 5I_1 - 2jI_1 - 3I_1 + 3I_2 = 0$

$(-I_1)(8 - 2j) + 3I_2 = -10\angle 30$

$(8 - 2j)\, I_1 - 3I_2 = 10\angle 30$ —— ①

Applying KVL to mesh 02

$-3(I_2 - I_1) - 5jI_2 - 5(I_2 - I_3) = 0$

$-3I_2 + 3I_1 - 5jI_2 - 5I_2 + 5I_3 = 0$

$+3I_1 - I_2(8 + j5) + 5I_3 = 0$ —— ②

Applying KVL to mesh 03

$-2I_3 - j2I_3 - 5(I_3 - I_2) = 0$

$-2I_3 - j2I_3 - 5I_3 + 5I_2 = 0$

$5I_2 - I_3(7 - j2) = 0$

using Cramer's Rule.

$$\begin{bmatrix} 8-j2 & -3 & 0 \\ -3 & 8+j5 & -5 \\ 0 & -5 & 7-j2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10\angle 30° \\ 0 \\ 0 \end{bmatrix}$$

$\Delta = (8 - 2j)\left[(8 + j5)(7 - 2j) - (-5 \times -5)\right] + 3\left[(-3)(7 - 2j)\right] + 0$

$\Delta = (8 - 2j)\left[66 + 19j - 25\right] + 3(-21 + 6j)$

$= (8 - 2j)(41 + 19j) - 63 + 18j$

$\Delta = 366 + 70j - 63 + 18j$

$\Delta = 303 + 18j = 315.52\angle 16.19$

$$\Delta 1 = \begin{bmatrix} 10\angle 30° & -3 & 0 \\ 0 & 8+j5 & -5 \\ 0 & -5 & 7-j2 \end{bmatrix}$$

$\Delta 1 = 10\angle 30° \left[41 + 19j\right] + 3(0) + 0$

$\Delta 1 = 10\angle 30° \times 45\angle 24.86$

$I_1 = \dfrac{\Delta 1}{\Delta} = \dfrac{10\angle 30 \times 45\angle 24.86}{315.52\angle 16.19}$

$\boxed{I_1 = 1.43\angle 38.67\ A}$

$$\Delta 2 = \begin{bmatrix} 8-j2 & 10\angle 30° & 0 \\ -3 & 0 & -5 \\ 0 & 0 & 7-j2 \end{bmatrix}$$

$= 0 - 10\angle 30° \left[-3(7 - 2j)\right] + 10$

$= -10\angle 30° \left[-21 + 6j\right]$

$\Delta 2 = -10\angle 30° \times 21.84\angle -15.95$

$I_2 = \dfrac{\Delta 2}{\Delta} = \dfrac{-10\angle 30° \times 21.84\angle -15.95}{315.52\angle 16.19}$

$\boxed{I_2 = 0.693\angle -2.2\ A}$

$$\Delta_3 = \begin{bmatrix} 8-2j & -3 & 10\angle 30^\circ \\ -3 & 8+j5 & 0 \\ 0 & -5 & 0 \end{bmatrix}$$

$\Delta_3 = 0 + 0 + 10\angle 30^\circ\,[+15 - 0]$

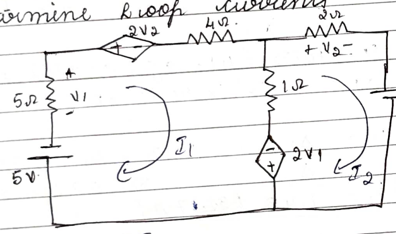
$= 10\angle 30^\circ\,(15)$

$\Delta_3 = 150\angle 30^\circ$

$I_3 = \dfrac{\Delta_3}{\Delta} = \dfrac{150\angle 30^\circ}{315.52\angle 16.19}$

$\boxed{I_3 = 0.476\angle 13.8\ A}$

4) Determine loop currents



=> $V_1 = -5I_1$

$V_2 = 2I_2$

Apply KVL to mesh 01.

$-5 - 5I_1 - 2V_2 - 4I_1 - (I_1 - I_2) + 2V_1 = 0$

$-5 - 5I_1 - 2(2I_2) - 4I_1 - I_1 + I_2 + 2V_1 = 0$

$20I_1 + 3I_2 = -5$ —— ①

Applying KVL to mesh 02.

$-2V_1 - (I_2 - I_1) - 2I_2 - 10 = 0$

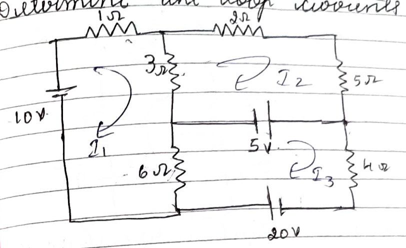
$-2(-5I_1) - I_2 + I_1 - 2I_2 = 10$

$11I_1 - 3I_2 = 10$ —— ②

Solving ① & ②

$\boxed{I_1 = 0.161 A}$ $\boxed{I_2 = -2.742 A}$

5) Determine the loop currents



=> Apply KVL to mesh 01

$10 - 1(I_1) - 3(I_1 - I_2) - 6(I_1 - I_3) = 0$

$10 - I_1 - 3I_1 + 3I_2 - 6I_1 + 6I_3 = 0$

$-10I_1 + 3I_2 + 6I_3 = -10$

$10I_1 - 3I_2 - 6I_3 = 10$ —— ①

Apply KVL to mesh 02

$-2I_2 - 5I_2 - 5 - 3(I_2 - I_1) = 0$

$-2I_2 - 5I_2 - 5 - 3I_2 + 3I_1 = 0$

$\cancel{-3I_1 + 10I_2}$ $-3I_1 + 10I_2 = 5$ —— ②

Apply KVL to mesh 03

$+5 - 4I_3 + 20 - 6(I_3 - I_1) = 0$

$5 - 4I_3 + 20 - 6I_3 + 6I_1 = 0$

$6I_1 - 10I_3 + 25 = 0$ —— ③

$-6I_1 + 10I_3 = 25$

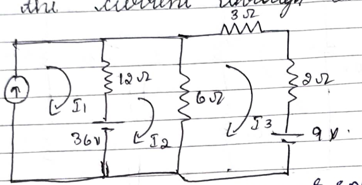
Solving ① ② ③

$\boxed{I_1 = 4.27 A}$

$\boxed{I_2 = 0.79 A}$

$\boxed{I_3 = 5.06 A}$

(6) Determine the loop currents and find the current through 2Ω resistor



=> Mesh 1 contains current source of 6A

∴ $\boxed{I_1 = 6A}$

Apply KVL to mesh 02

$36 - 12(I_2 - I_1) - (I_2 - I_3)6 = 0$

$36 - 12I_2 + 12I_1 - 6I_2 + 6I_3 = 0$

$36 - 18I_2 + 12(6) + 6I_3 = 0$

$-18I_2 + 6I_3 = -108$

$18I_2 - 6I_3 = 108$ —— (1)

Apply KVL to mesh 03

$-3I_3 - 2I_3 - 9 - 6(I_3 - I_2) = 0$

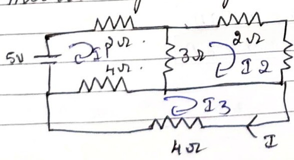
$-3I_3 - 2I_3 - 9 - 6I_3 + 6I_2 = 0$

$6I_2 - 11I_3 = 9$ —— (2)

Solving (1) & (2)

$\boxed{I_2 = 3A}$ $\boxed{I_3 = 3A}$

(7) Determine the loop currents using loop method of analysis



=> ~~Apply KVL for mesh 01~~

$\cancel{5 - 2I_1 - 3(I_1 - I_2) - 4(I_1 - I_3) = 0}$

$\cancel{5 - 2I_1 - 3I_1 + 3I_2 - 4I_1 + 4I_3 = 0}$

$\cancel{-9I_1 + 7I_2 + 4I_3 = 5}$

$\cancel{4I_3 + 9I_1 - 7I_2 = +5}$ —— (1)

Apply KVL to mesh 02

$-3(I_2 - I_1) - 2I_2 - 3I_2 = 0$

$-3I_2 + 3I_1 - 2I_2 - 3I_2 = 0$

$\cancel{-3I_1 + 8I_2 = 0}$

$-3I_1 + 8I_2 = 0$ —— (1)

Apply KVL to mesh 03

$-4(I_3 - I_1) - 4I_3 = 0$

$-4I_3 + 4I_1 - 4I_3 = 0$

$-4I_1 + 8I_3 = 0$ —— (2)

Apply KVL to mesh 01

$5 - 2I_1 - 3(I_1 - I_2) - 4(I_1 - I_3) = 0$

$5 - 2I_1 - 3I_1 + 3I_2 - 4I_1 + 4I_3 = 0$

$-9I_1 + 3I_2 + 4I_3 = -5$

$9I_1 - 3I_2 - 4I_3 = 5$ —— (3)

$\boxed{I_1 = 0.851}$

$\boxed{I_2 = 0.319}$

$\boxed{I_3 = 0.425}$

(8) Explain the classification of electrical network

=> The Electrical networks are classified into

a) Unilateral (or) Bilateral

(b) Linear (or) nonlinear
(c) Lumped (or) distributed
(d) Active (or) Passive

a) Unilateral (or) Bilateral n/w:
Unilateral n/w is the one whose properties / characteristics ~~do not~~ change ~~to~~ with direction of operation.
Eg: diode, rectifier

Bilateral n/w is the one whose properties / characteristics do not change / are same in either direction.
Eg: Resistance.

b) Linear (or) nonlinear n/w:
Linear n/w is the one whose parameters are constant i.e., they do not change with voltage (or) current. It obeys law of superposition.
Eg: ~~Mesh~~ Mesh.

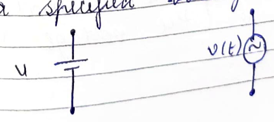
Nonlinear n/w is the one whose ~~properties~~ parameters change with voltage or current (It does not obey law of superposition)
Eg: Electrical elements

c) Lumped (or) distributed:
Lumped n/w is the one in which various electrical elements can be separable physically and it is represented by ordinary differential eqn.
Eg: RLC.
A distributed n/w is the one in which various electrical elements cannot be seperated physically.
Eg: Power transmission lines.

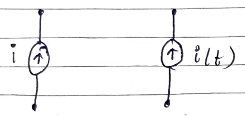
d) Active (or) Passive:-
Active n/w is the one in which consists of one (or) more emf source in it.
Eg: voltage source, current source.
Passive n/w is the one in which contains no source of emf.
Eg: resistor, inductor.

(9) Explain with an example dependent independent sources and supermesh.

=> InDependent Sources:-
O/P characteristics of an independent source are not dependent on any n/w variable such as a current or voltage. It is of 2 types
i) Independent voltage source: It is the one which ~~has~~ is a two terminal n/w element that establishes a specified voltage across its terminals



2) Independent current source: It is the one which is a two terminal n/w element which produces a specified current.

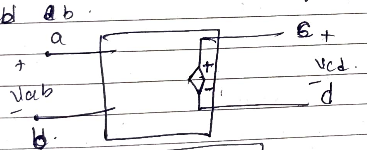


Dependent Sources:-

If the voltage or current of a source depends in turn upon some other voltage or current it is called as dependent sources.

a) Voltage Controlled Voltage Source [VCVS]

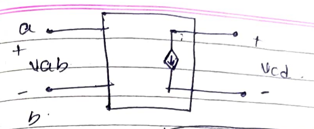
It is a four terminal n/w that establishes a voltage Vcd. b/w two points c and d in the ckt. that is propotional to a voltage Vab b/w a and b.



$\boxed{V_{cd} = \mu V_{ab}}$

b) Voltage Controlled Current Source [VCCS]

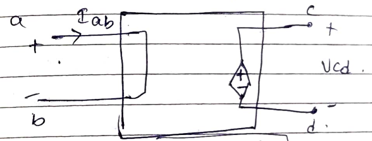
It is a four terminal m/w component that establishes current icd in a branch of the ckt that is propotional to the voltage Vab b/w a and b.



$\boxed{i_{cd} = g_m V_{ab}}$

c) ~~cont~~ Current Controlled Voltage Source [VCVS]

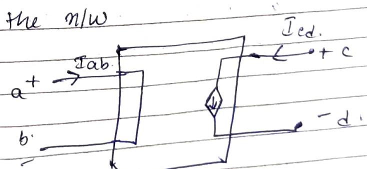
It is a four terminal n/w component that establishes voltage Vcd b/w two points c and d in the ckt that is propotional to current Iab in some branch of the ckt.



$\boxed{V_{cd} = r I_{ab}}$

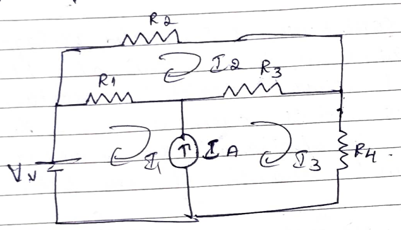
d) Current Controlled Current Source [CCCS]

It is the four terminal n/w component that establishes a current Icd in one branch of a ckt that is propotional to the current Iab in some branch of the n/w

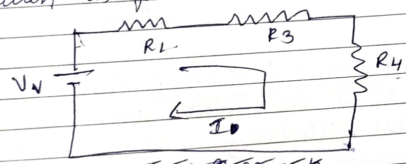


$\boxed{I_{cd} = \beta I_{ab}}$

Supermesh :- Meshes that shares a current source with other meshes, none of which contains a current source in the outer loop, form a ~~super mesh~~ supermesh



In the above n/w I_1 & I_3 forms supermesh and the n/w can be written as follows



~~The supermesh eqn can b~~ loop eq^n for supermesh is

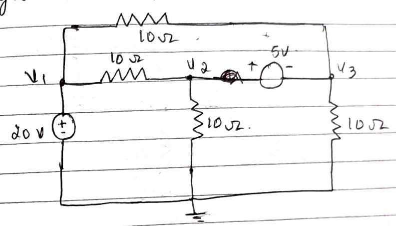
$V - R_1(I_1 - I_2) - R_3(I_3 - I_2) - R_4 I_3 = 0$

The constrait eq^n can be written as

$\boxed{I_3 - I_1 = I}$

Unit (02) :-

① Apply nodal Analysis to find node voltages V_1, V_2 and V_3.



=> Since V_1 is directly connected to 20V source $\boxed{V_1 = 20V}$

V_2 and V_3 forms supernode

Apply KCL to supernode

$$\frac{V_2 - V_1}{10} + \frac{V_2}{10} + \frac{V_3}{10} + \frac{V_3 - V_1}{10} = 0$$

$$\frac{V_2 - 20}{10} + \frac{V_2}{10} + \frac{V_3}{10} + \frac{V_3 - 20}{10} = 0$$

X^{ly} by 10 on BS

$(V_2 - 20) + V_2 + V_3 + (V_3 - 20) = 0$

$2V_2 + 2V_3 - 40 = 0$

$2V_2 + 2V_3 = 40$

$V_2 + V_3 = 20$ —— ①

The voltage constraint eq^n

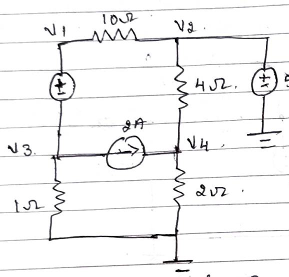
$V_2 - V_3 = 5$ —— ②

Solving ① & ②

$\boxed{V_2 = 12.5V}$

$\boxed{V_3 = 7.5V}$

(2) For the circuit shown in figure determine all the node voltages.



=> node 1 and 3 forms supernode.

constraint eq^n $V_1 - V_3 = 6$ —— (1)

Apply KCL to ~~node~~ supernode.

$$\frac{V_1 - V_2}{10} + \frac{V_3}{1} + 2 = 0$$

Since V_2 is directly connected to 5V source

$$\boxed{V_2 = 5V}$$

$$\frac{V_1 - 5}{10} + \frac{V_3}{1} + 2 = 0$$

$$0.1 V_1 + V_3 = -1.5 \text{ —— (2)}$$

Apply KCL to node 4

$$\frac{V_4 - V_2}{4} + \frac{V_4}{2} = 2$$

$$0.75 V_4 = 2 + \frac{V_2}{4}$$

$$0.75 V_4 = 3.25$$

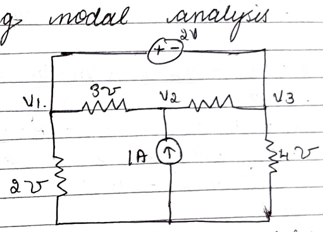
$$V_4 = \frac{3.25}{0.75}$$

$$\boxed{V_4 = 4.33 V}$$

Solving (1) & (2)

$\boxed{V_1 = 4.09 V}$ $\boxed{V_3 = -1.90 V}$

(3) Determine all the node voltages using nodal analysis



=> V_1 and V_3 forms supernode.

constraint voltage eq^n

$$V_1 - V_3 = 2V \text{ —— (1)}$$

Apply KCL to ~~node~~ supernode.

$$3(V_1 - V_2) + (V_1 - 0)2 + (V_3 - V_2)2 + (V_3 - 0)4 = 0$$

~~Apply~~ $3V_1 - 3V_2 + V_1 2 + 2V_3 - 2V_2 + 4V_3 = 0$

$$5V_1 - 5V_2 + 6V_3 = 0 \text{ —— (2)}$$

Apply KCL to node 2

$$3(V_2 - V_1) + (V_2 - V_3)2 = 1$$

$$3V_2 - 3V_1 + 2V_2 - 2V_3 = 1$$

$$-3V_1 + 5V_2 - 2V_3 = 1 \text{ —— (3)}$$

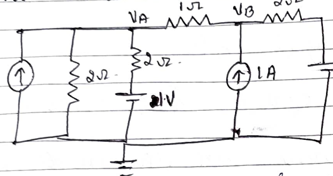
Solving (1) & (2) & (3)

$\boxed{V_3 = 0.83 V}$

$\boxed{V_1 = -1.16 V}$

$\boxed{V_2 = -0.16 V}$

4) Find V_A and V_B.



=> Apply KCL at node V_A.

$$-2 + \frac{V_A}{2} + \frac{V_A-1}{2} + \frac{V_A-V_B}{1} = 0$$

$$-2 + V_A\left[\frac{1}{2}+\frac{1}{2}+1\right] - V_B - \frac{1}{2} = 0$$

$$-\frac{5}{2} + V_A[2] - V_B = 0$$

$$2V_A - V_B = 5/2$$

x'ly by 2 on BS.

$$4V_A - 2V_B = 5 \quad \text{—— (1)}$$

Apply KCL to node V_B.

$$\frac{V_B-V_A}{1} + \frac{V_B-2}{2} - 1 = 0$$

$$V_B[1-1/2] - V_A - 1 - 1 = 0$$

$$\frac{3V_B}{2} - V_A - 2 = 0$$

x'ly by 2.

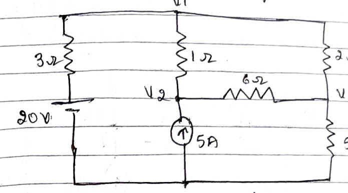
$$3V_B - 2V_A - 4 = 0$$

$$-2V_A + 3V_B = 4 \quad \text{—— (2)}$$

Solving (1) & (2)

$\boxed{V_A = 2.875\,V}$ $\boxed{V_B = 3.25\,V}$

(5) Determine the node voltages using nodal method of analysis.



Apply KCL at node 1

$$\frac{V_1-V_2}{1} + \frac{V_1-20}{3} + \frac{V_1-V_3}{2} = 0$$

$$\left(\frac{1}{3}+1+\frac{1}{2}\right)V_1 - V_2 - \frac{V_3}{2} - \frac{20}{3} = 0$$

$$1.83V_1 - V_2 - 0.5V_3 = \frac{20}{3}$$

$$1.83V_1 - V_2 - 0.5V_3 = 6.67 \quad \text{—— (1)}$$

Apply KCL at node 2.

$$\frac{V_2-V_1}{1} + \frac{V_2-V_3}{6} - 5 = 0$$

$$\left(1+\frac{1}{6}\right)V_2 - V_1 - \frac{V_3}{6} = 5$$

$$-V_1 + 1.17V_2 - 0.17V_3 = 5 \quad \text{—— (2)}$$

Apply KCL at node 3

$$\frac{V_3-V_2}{6} + \frac{V_3-V_1}{2} + \frac{V_3}{5} = 0$$

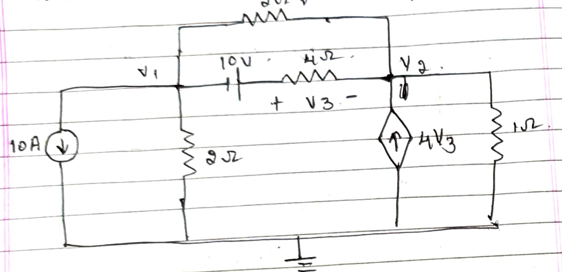
$$\left(\frac{1}{6}+\frac{1}{2}+\frac{1}{5}\right)V_3 - \frac{V_2}{6} - \frac{V_1}{2} = 0$$

$$-0.5V_1 - 0.17V_2 + 0.87V_3 = 0 \quad \text{—— (3)}$$

Solving (1), (2) & (3)

$\boxed{V_1 = 23.82\,V}$ $\boxed{V_2 = 27.4\,V}$ $\boxed{V_3 = 19.04\,V}$

(6) Determine the node voltages using nodal method of analysis



=> Apply KCL to node 1.

$$10 + \frac{V_1}{2} + \frac{V_1 - V_2}{2} + \frac{V_1 + 10 - V_2}{4} = 0$$

$$\left(\frac{1}{2} + \frac{1}{2} + \frac{1}{4}\right) V_1 - \left(\frac{1}{4} + \frac{1}{2}\right) V_2 = -10 - \frac{10}{4}$$

$$1.25V_1 - 0.75V_2 = -12.5 \quad \text{—(1)}$$

Apply KCL to node 2.

$$\frac{V_2 - 10 - V_1}{4} + \frac{V_2 - V_1}{2} + \frac{V_2}{1} = 4V_3$$

$$\frac{V_2 - 10 - V_1}{4} + \frac{V_2 - V_1}{2} + \frac{V_2}{1} = 4V_3$$

WKT

$$V_3 = V_1 - (-10) - V_2 = V_1 + 10 - V_2$$

$$\frac{V_2 - 10 - V_1}{4} + \frac{V_2 - V_1}{2} + \frac{V_2}{1} = 4(V_1 + 10 - V_2)$$

$$\frac{V_2 - 10 - V_1}{4} + \frac{V_2 - V_1}{2} + \frac{V_2}{1} - V_1 4 + 4V_2 = 40$$

$$V_2\left(\frac{1}{4} + \frac{1}{2} + 1 + 4\right) - V_1\left(\frac{1}{4} + \frac{1}{2} + 4\right) = 40 + \frac{10}{4}$$

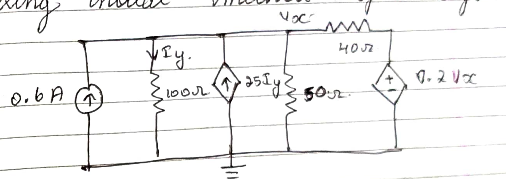
$$-4.75V_1 + 5.75V_2 = 42.5 \quad \text{—(2)}$$

Solving (1) & (2)

$\boxed{V_1 = -11.03\,V}$

$\boxed{V_2 = -1.72\,V}$

(7) Determine the node voltage Vx using nodal method of analysis



=> Apply KCL at node Vx.

~~25Iy + 0.6~~

$$\frac{V_x}{100} + \frac{V_x}{50} + \frac{V_x - 0.2V_x}{40} - 0.6 - 25I_y = 0$$

W.K.T

$$I_y = \frac{V_x}{100}$$

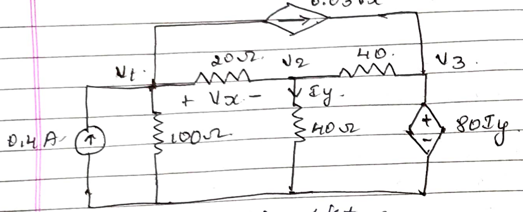
$$\frac{V_x}{100} + \frac{V_x}{50} + \frac{0.8V_x}{40} - 0.6 - 25\left(\frac{V_x}{100}\right) = 0$$

$$V_x\left(\frac{1}{100} + \frac{1}{50} + \frac{0.8}{40} - \frac{25}{100}\right) = 0.6$$

$$V_x(-0.2) = 0.6$$

$\boxed{V_x = -3\,V}$

(8) Determine the node voltages using Nodal method of analysis.



⇒ From the ckt.

$V_x = V_1 - V_2$

$I_y = \frac{V_2}{40}$

Apply KCL to node V_1.

$\frac{V_1}{100} + \frac{V_1 - V_2}{20} + 0.03V_x - 0.4 = 0$

$\frac{V_1}{100} + \frac{V_1 - V_2}{20} + 0.03(V_1 - V_2) - 0.4 = 0$

$V_1\left(\frac{1}{100} + \frac{1}{20} + 0.03\right) - \left(\frac{1}{20} + 0.03\right)V_2 = 0.4$

$0.09V_1 - 0.08V_2 = 0.4$ —— (1)

Apply KCL to node 2

$\frac{V_2 - V_1}{20} + \frac{V_2 - V_3}{40} + \frac{V_2}{40} = 0$

$V_2\left(\frac{1}{20} + \frac{1}{40} + \frac{1}{40}\right) - \frac{V_1}{20} - \frac{V_3}{40} = 0$

$-0.05V_1 + 0.1V_2 - 0.025V_3 = 0$ —— (2)

Apply KCL at node 3 since V_3 is directly connected to $80I_y$ source

$V_3 = 80I_y$.

$V_3 = 80\left(\frac{V_2}{40}\right)$

$V_3 = 2V_2$.

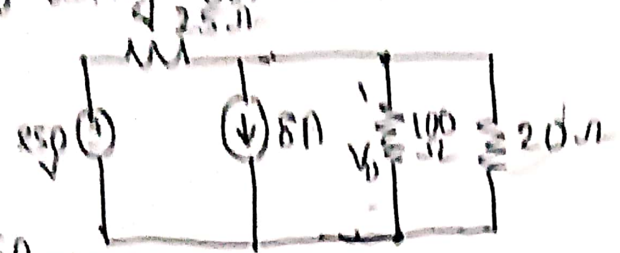
$2V_2 - V_3 = 0$ —— (3)

Solving (1) (2) (3).

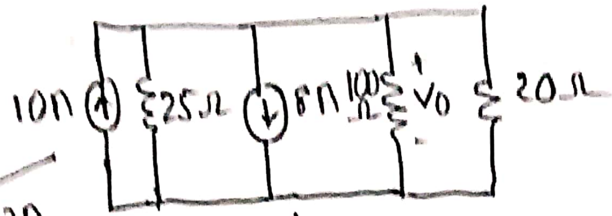
$\boxed{V_1 = 40V}$ $\boxed{V_2 = 40V}$ $\boxed{V_3 = 80V}$

Network Analysis
Module 1 Unit 3

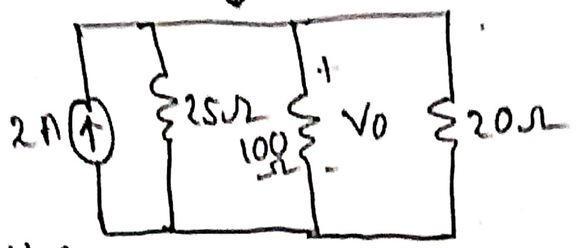
(1) Find V_0 using source Transformation?



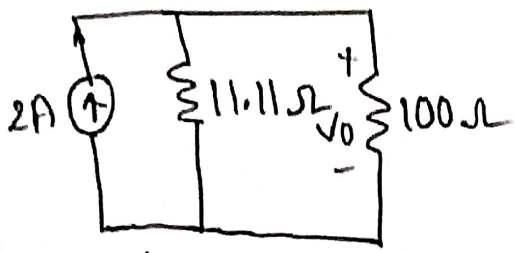
$I = \frac{V}{R} = \frac{250}{25} = 10A$



$10A - 8A = 2A$



$25 \| 20 = 11.11\Omega$



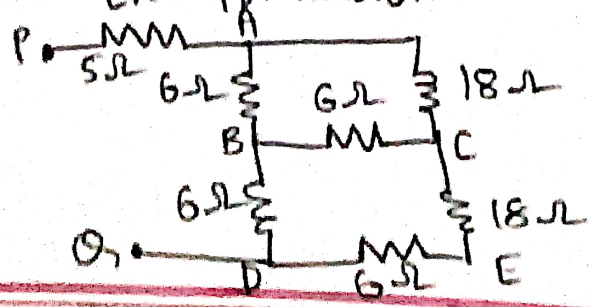
$$\frac{1}{R_{Total}} = \frac{1}{11.11} + \frac{1}{100}$$

$$R_{Total} = 10\Omega$$

$$V_0 = I_{net} \times R_{Total}$$

$$= 2 \times 10\Omega = 20V$$

(2) Determine the input resistance between PQ using star-delta transformation.



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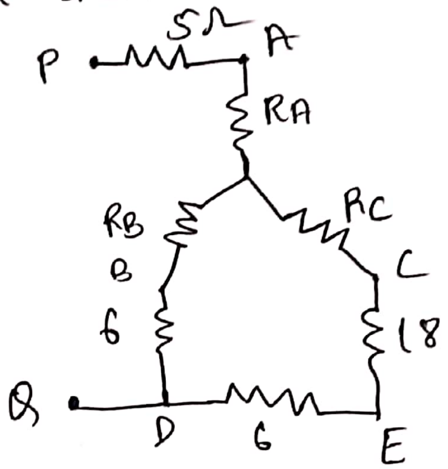
ABC forms a closed delta loop.

$R_{AB} = 6\Omega$

$R_{BC} = 6\Omega$ By formulae $R_{AB1} = \frac{R_A R_B}{R_A + R_B + R_C}$ & others

$R_{AC} = 18\Omega$

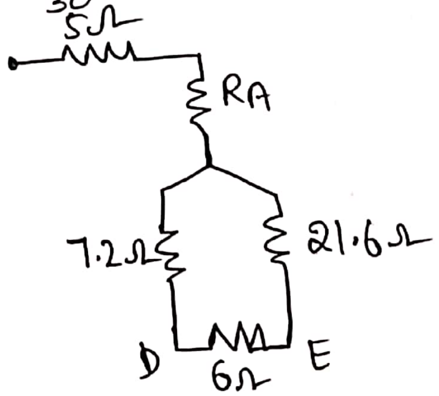


$R_{sum} = R_{AB} + R_{BC} + R_{AC} = 6 + 6 + 18 = 30\Omega$

$R_A = \frac{R_{AB} \times R_{AC}}{sum} = \frac{6 \times 18}{30} = 3.6\Omega$

$R_B = \frac{R_{AB} \times R_{BC}}{sum} = \frac{6 \times 6}{30} = 1.2\Omega$

$R_C = \frac{R_{BC} \times R_{AC}}{sum} = \frac{6 \times 18}{30} = 3.6\Omega$



$R_{DNE} = R_{left} + R_{right} = 7.2 + 21.6 = 28.8\Omega$

$R_{DE} = \frac{28.8 \times 6}{28.8 + 6} = \frac{172.8}{34.8} = 4.965\Omega$

$R_{PQ} = 5\Omega + R_A + R_{DE}$

$= 5 + 3.6 + 4.965 = 13.565\Omega$

③ For the ckt shown in fig. Determine equivalent resistance between any 2 terminals

2 n/w → outer delta n/w
inner star n/w

convert inner star n/w to delta

$R_x = 4\Omega$, $R_y = 4\Omega$, $R_z = 4\Omega$

$R_A = R_y + R_z + \frac{R_y R_z}{R_x} = 12$

$R_B = 12$
$R_C = 12$ } similarly

Now
Inner delta branch countaining 12Ω at 3 sides is in parallel with Outer delta branch countaining 4Ω

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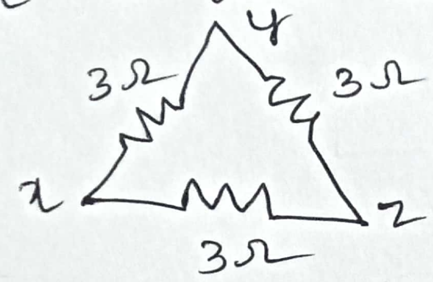
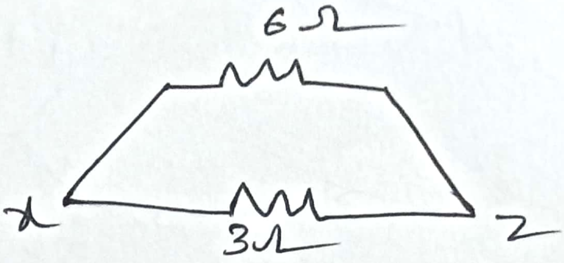
$\therefore \quad R_P = \frac{4\times12}{4+12} = \frac{48}{16} = 3\Omega$

entire ckt is simplified with 3Ω on each side a delta loop





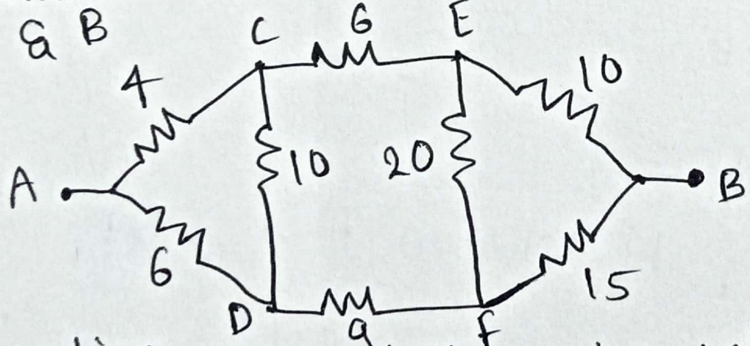
x-y & y-z are in series

Now by last n/w

$6 \| 3 = \frac{3\times6}{3+6} = \frac{\cancel{18}^{2}}{\cancel{9}_{1}} = 2\Omega$

$R_{eq} = 2\Omega$

4 Determine the equivalent resistance between terminals A & B



converting left delta n/w into star n/w

$R_{AC} = 4\Omega$

$R_{AD} = 6\Omega$

$R_{CD} = 10\Omega$

sum $= 20\Omega$

$R_A = \frac{4\times6}{4+6+10} = \frac{24}{20} = 1.2\Omega$

$R_B = \frac{4\times10}{20} = \frac{40}{20} = 2\Omega$

$R_D = \frac{6\times10}{20} = \frac{60}{20} = 3\Omega$

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converting right delta n/w into star

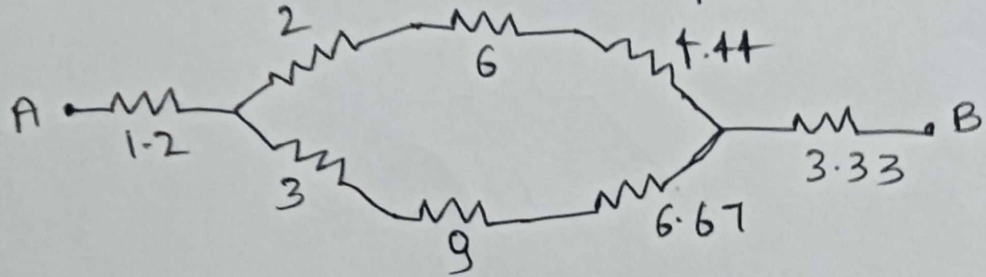
$R_{EB} = 10\Omega$ sum = $10+15+20 = 45\Omega$

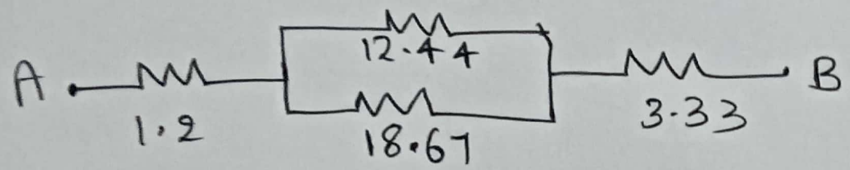
$R_{FB} = 15\Omega$ $R_B = \frac{10\times15}{45} = \frac{150}{45} = 3.33\Omega$

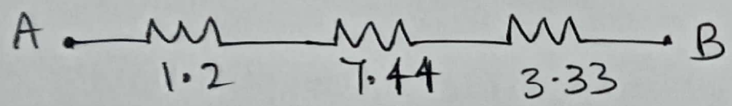
$R_{EF} = 20\Omega$

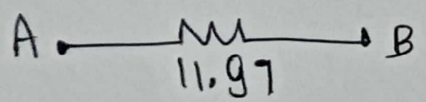
$R_E = \frac{10\times20}{45} = 4.44\Omega$

$R_F = \frac{15\times20}{45} = 6.67\Omega$



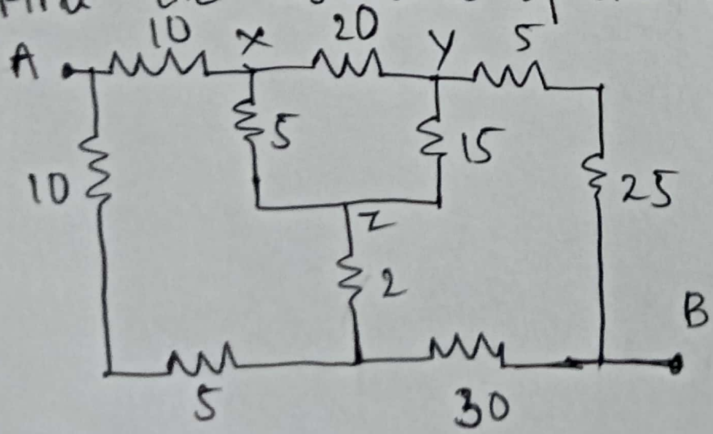






$R_{AB} = 11.97 \approx 12\Omega$

5) Find the resistance of ckt shown b/w A & B



$R_x = \frac{R_{xy} \times R_{xz}}{sum} = \frac{20\times5}{40} = 2.5\Omega$

$R_y = \frac{R_{xy} \times R_{yz}}{sum} = \frac{20\times15}{40} = 7.5\Omega$

$R_z = \frac{R_{xz} \times R_{yz}}{sum} = \frac{5\times15}{40} = 1.875\Omega$

$R_{xy} = 20\Omega$

$R_{xz} = 5\Omega$

$R_{yz} = 15\Omega$

sum = $20+5+15 = 40\Omega$

$\longrightarrow$

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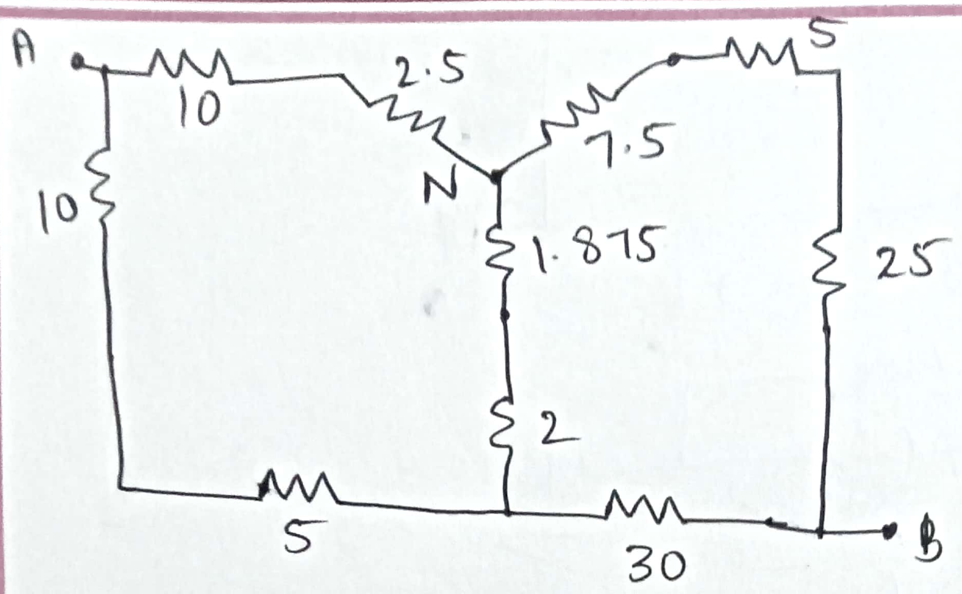
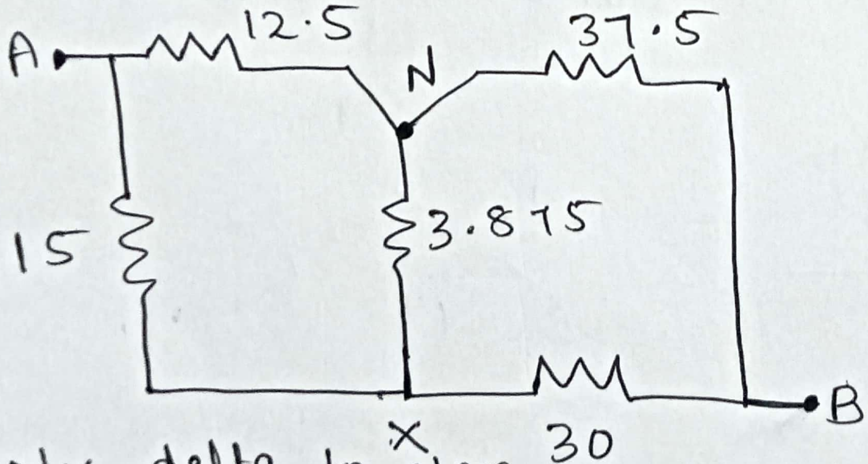


Now → $10+2.5=12.5\,\Omega$ → R_{branch_x}

R_{branch_y} → $30+7.5=37.5\,\Omega$

R_{branch_z} → $1.875+2=3.875\,\Omega$



ANX - delta to star

$$R_1=\frac{12.5\times 3.875}{31.37}=1.54\,\Omega$$

$$R_2=\frac{12.5\times 15}{31.37}=5.97\,\Omega$$

$$R_3=\frac{3.875\times 15}{31.37}=1.85$$

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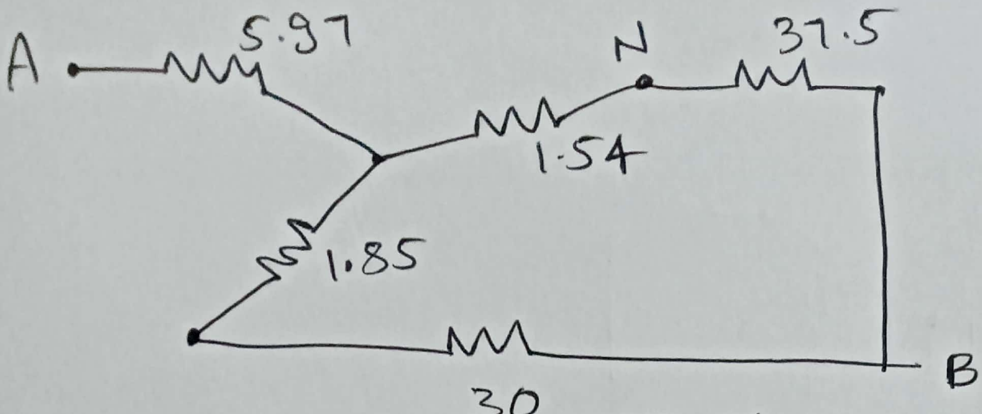
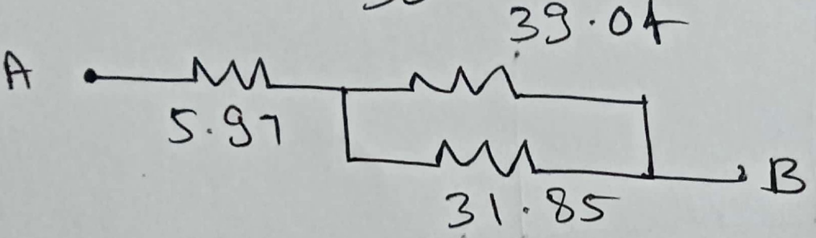
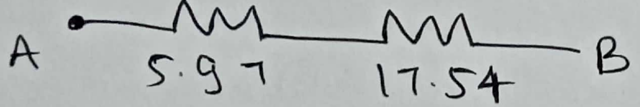
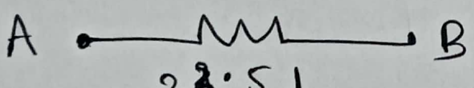
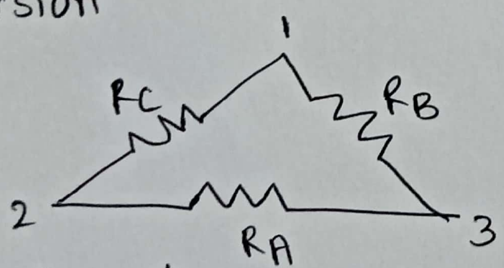


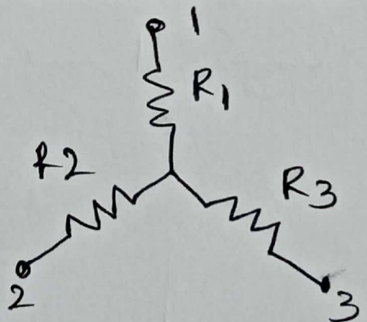






(6) obtain the expression for star to delta and delta to star conversion





Delta to star Transformation

From terminal 1 & 2 $= R_C \parallel (R_A + R_B) = \dfrac{R_C(R_A+R_B)}{R_C+R_A+R_B}$

Referring to star n/w shown the resistance b/w 1 & 2 $= R_1 + R_2$.

since 2 ntw are electrically equivalent

$$R_1 + R_2 = \frac{R_C(R_A+R_B)}{R_A+R_B+R_C} \quad (1)$$

$$\& \; R_3 + R_1 = \frac{R_B(R_A+R_C)}{R_A+R_B+R_C} \quad (3)$$

similarly,

$$R_2 + R_3 = \frac{R_A(R_B+R_C)}{R_A+R_B+R_C} \quad (2)$$

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substracting ② & ①

$$R_1 - R_3 = \frac{R_B R_C - R_A R_B}{R_A + R_B + R_C} \quad \text{—④}$$

Add ④ & ③

$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C}$$

similarly,

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_A R_B}{R_A + R_B + R_C}$$

Thus, star resistor connected to a terminal is equal to product of 2 delta networks connected to same terminal divided by sum of delta ~~nf~~ resistors.

star to delta Transformation.

multiplying above eq

$$R_1 R_2 = \frac{R_A R_B R_C^2}{(R_A + R_B + R_C)^2} \quad \text{—⑤}$$

$$R_2 R_3 = \frac{R_A^2 R_B R_C}{(R_A + R_B + R_C)^2} \quad \text{—⑥}$$

$$R_3 R_1 = \frac{R_A R_B^2 R_C}{(R_A + R_B + R_C)^2} \quad \text{—⑦}$$

Add ⑤, ⑥, ⑦

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$R_1R_2 + R_2R_3 + R_3R_1$

$$= \frac{R_A R_B R_C (R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_A R_B R_C}{R_A + R_B + R_C}$$

$$= R_A R_1 = R_B R_2 = R_C R_3$$

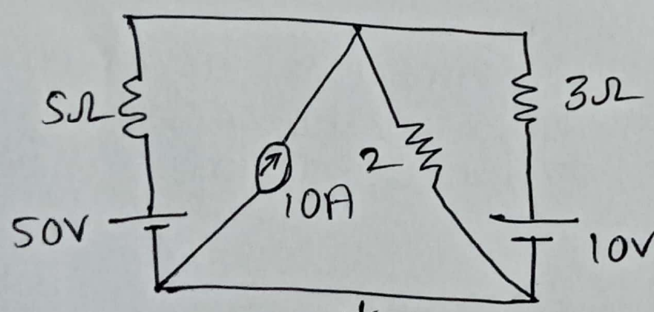
Hence,

$$R_A = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1} = R_2 + R_3 + \frac{R_2R_3}{R_1}$$

$$R_B = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2} = R_1 + R_3 + \frac{R_3R_1}{R_2}$$

$$R_C = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3} = R_1 + R_2 + \frac{R_1R_2}{R_3}$$

7) Determine power delivered by 50V source using source transformation

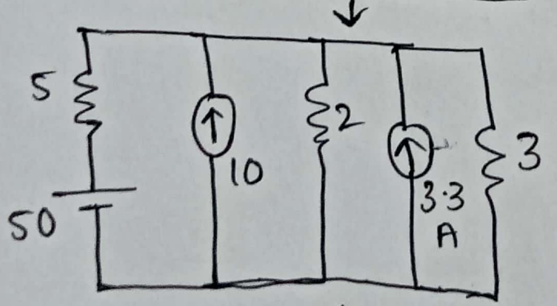


$V = IR$

$I = V/R$

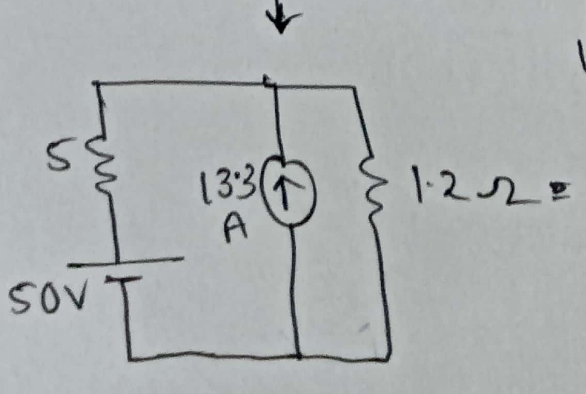
$= \frac{10}{3}$

$= 3.33 A$

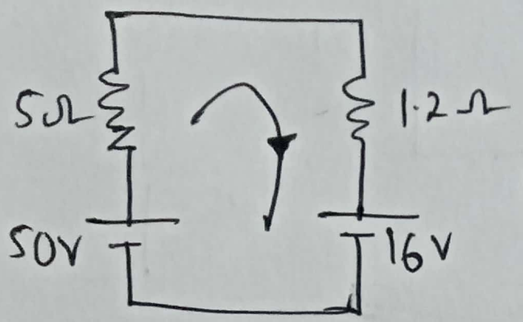


$2\Omega \| 3\Omega = 1.2\Omega$

$10 + 3.3 = 13.3 A$



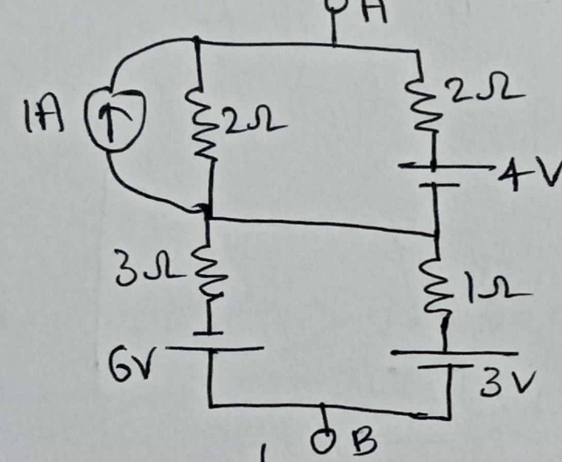
By source Transformation:

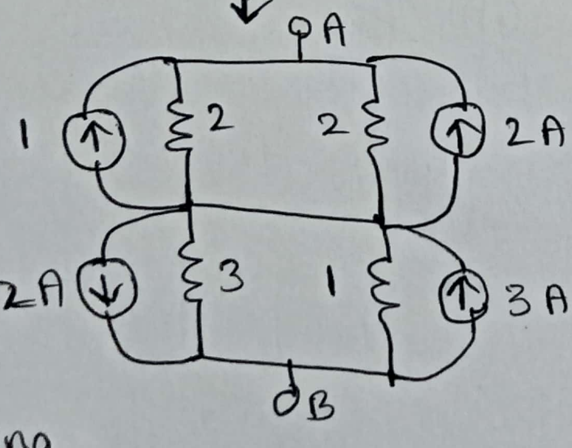


$$I = \frac{50-16}{5+1.2} = 5.48\,A$$

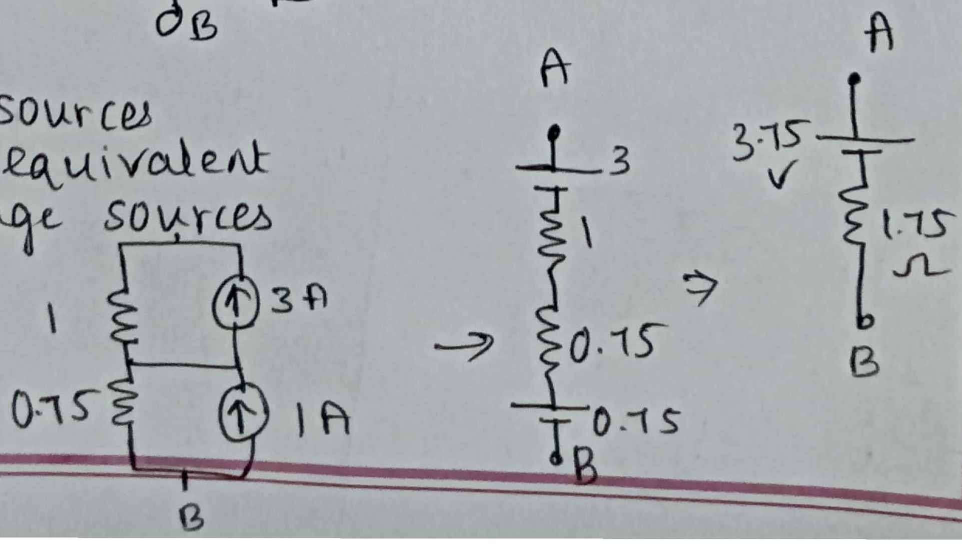
$$Power = 50 \times 5.48 = 274\,W$$

(8) Reduce the n/w shown in to a single source n/w





converting current sources into equivalent voltage sources



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